

Introduction to AI Fairness

Yunfeng Zhang
zhangyun@us.ibm.com
IBM Research AI
Yorktown Heights, NY, USA

Q. Vera Liao
vera.liao@ibm.com
IBM Research AI
Yorktown Heights, NY, USA

Rachel K. E. Bellamy
IBM Research AI
Yorktown Heights, NY, USA
rachel@us.ibm.com

Moninder Singh
moninder@us.ibm.com
IBM Research AI
Yorktown Heights, NY, USA

ABSTRACT

Today, AI is used in many high-stakes decision-making applications in which fairness is an important concern. Already, there are many examples of AI being biased and making questionable and unfair decisions. Recently, the AI research community has proposed many methods to measure and mitigate unwanted biases, and developed open-source toolkits for developers to make fair AI. This course will cover the recent development in algorithmic fairness, including the many different definitions of fairness, their corresponding quantitative measurements, and ways to mitigate biases. This course is open to beginners and is designed for anyone interested in the topic of AI fairness.

CCS CONCEPTS

• **Social and professional topics** → **Computing / technology policy**; • **Computing methodologies** → **Machine learning**.

KEYWORDS

algorithmic fairness, bias, decision support, discrimination-aware machine learning

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1 INTRODUCTION

Machine learning models are increasingly used to inform high-stake decisions about people. Although machine learning, by its very nature, is always a form of statistical discrimination, the discrimination becomes objectionable when it places certain privileged groups at systematic advantage and certain unprivileged groups at systematic disadvantages. Recently, the AI research community began addressing AI biases by introducing metrics to measure fairness of AI predictions and algorithms to mitigate algorithmic biases

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(e.g., [2, 5]). However, we believe this issue would benefit a lot from the attention of the HCI research community, as there are still many unknowns about the human needs on fairness and whether the algorithmic definition of fairness actually map to various stakeholders' perception of fairness.

This course is intended for HCI researchers and practitioners who are interested in developing and designing fair AI systems, and those who want to understand the trends and core topics in the AI fairness literature. The course will also introduce AIF360, a comprehensive open source toolkit providing implementations of the metrics and algorithms developed by the broader algorithmic fairness research community.

2 BENEFITS

Attendees of the course will benefit from:

- Having an overview of the concepts, trends and core topics in the field of algorithmic fairness research
- Gaining an understanding of various definitions of fairness and how they can be measured
- Learning about the availability of open-source toolkits to support measuring and improving AI fairness
- Learning about the AIF360 toolkit [1] in particular and how to choose metrics and mitigation algorithms from the toolkit
- Hands-on experience in using the toolkit to assess model bias and correct it.

3 INTENDED AUDIENCE AND PREREQUISITES

The intended audience for this course is any CHI attendees who have already, or intend to engage in developing, designing and researching on the topic of AI fairness. The course does not require any advanced knowledge in AI, data science or programming, though a basic understanding of machine learning concepts such as classification, training data, and features could be helpful. The course will include 20-30 minutes hands-on practice with Python scripts provided. Interested attendees could further explore the code but programming is not required. Course instructors will provide introductory materials for machine learning and Python programming beforehand for interested attendees.

4 CONTENT

The course will be divided into three parts:

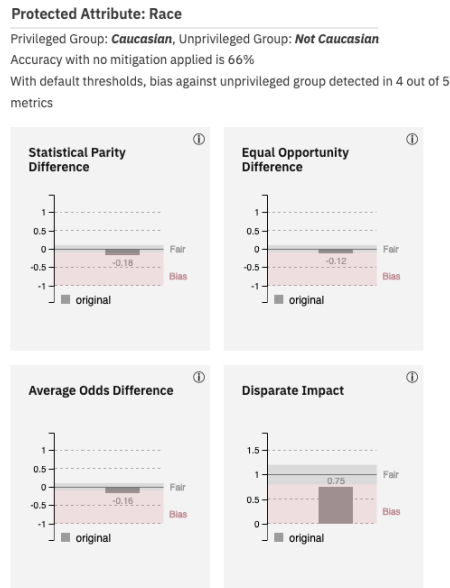


Figure 1: A screenshot of bias checking visualizations shown on the AIF360 web site.

4.1 Part 1: Overview of AI fairness issues (30 minutes)

The course will begin with an introductory definition of the field of algorithmic fairness. We will discuss the application domains in which AI fairness issues often emerge, and introduce a common vocabulary used when discussing these issues.

We will discuss in detail the many definitions of fairness and how they map to different mathematical, social, legal and philosophical perspective. We will talk about why there are so many different definitions and how they may even conflict with each other [3, 4].

4.2 Part 2: Introduction to the AI fairness metrics and toolkits (30 minutes)

Having exposed the different metrics, we will introduce a list of open source toolkits that can be used to assess AI fairness. Among them, we will discuss AIF360 in detail because it is one of the few toolkits that implement not only the metrics but also various bias mitigation algorithms. Three of the authors are also involved in the development of this toolkit.

We will use the interactive demo website for AIF360 to discuss and demonstrate some of the mitigation algorithms applied to criminal justice and financial scenarios (e.g., see Figure 1). To help confront the potentially bewildering choice of metric that needs to be made in a given application, we will also provide some practical guidance for choosing fairness metrics and mitigation algorithms.

4.3 Part 3: Hands-on experience (20 minutes)

We will take the audience through a Jupyter Notebook that contains an example of applying AIF360 to a medical dataset. This example includes the whole process of building a fair classifier, from defining protected attributes, building a classifier and examining its fairness,

to bias mitigation. This example should provide the audience the necessary experience to apply AIF360 to their own applications.

5 PRACTICAL WORK

The audience can follow along with the hands on experience section using the provided Jupyter Notebook and the installation instructions. The audience can choose to change the Python code to try out different fairness metrics and bias mitigation methods.

6 INSTRUCTOR BACKGROUND

6.1 Yunfeng Zhang

Yunfeng is a Research Staff Member at IBM Research AI. His research interests lie in the intersection between HCI and AI. His recent research projects involved creating conversational agents, modeling social interactions, and studying AI explainability, fairness, and trust. He contributed to the IBM's AI Fairness 360 and AI Explainability 360 open source toolkits, which are designed to help AI developers create intelligible and fair AI solutions. He received his Ph.D. in computer and information science from the University of Oregon in 2015.

6.2 Rachel K. E. Bellamy

Rachel is a Principle Research Scientist and Chair of the Computer Sciences Council at IBM T J Watson Research Center, Yorktown Heights, New York. In this role she heads a Council that manages a Research portfolio of exploratory science projects. Prior to this, she led an interdisciplinary team of human-computer interaction researchers, user experience designers and software engineers. That team most recently contributed to several IBM Research's Trusted AI projects, including the AI Fairness 360 and AI Explainability 360. Rachel received her doctorate in cognitive psychology from University of Cambridge, UK in 1991. She received a Bachelor of Science in psychology with mathematics and computer science from University of London in 1986. Before coming to IBM Research, she worked in Apple Computer's Advanced Technology Group researching software support for collaborative learning.

6.3 Q. Vera Liao

Vera is a Research Staff Member in IBM Research AI. Her research interests are on human-AI interaction and intelligent user interfaces. Her current work focuses on user-aware conversational agents, human centered machine learning, AI explainability and fairness. Her work contributed to IBM's AI Explainability 360, an open-source toolkit providing educational and technical resources for AI explainability. Recently, she organized several workshops and a panel on topics that connect the HCI and AI communities at premier HCI conferences including ACM CSCW and IUI. She received her Ph.D. in computer science from the University of Illinois at Urbana-Champaign.

6.4 Moninder Singh

Moninder is a Research Staff Member in the IBM Research AI organization at the IBM T. J. Watson Research Center. He received his Ph.D. in Computer and Information Science from the University of Pennsylvania in 1998. He is primarily interested in developing

and deploying solutions for interesting analytics and decision support problems. His main research areas are machine learning and data mining, artificial intelligence, data privacy, information retrieval, probabilistic modeling and reasoning, and text mining. He has been actively working in issues of fairness and trust in AI, has contributed to the IBM's AI Fairness 360 and AI Explainability 360 open source toolkits, and has given several tutorials/talks and published papers on issues relating to trust in AI models.

7 RESOURCES

The main website accompanying this course is: <https://aif360.mybluemix.net>. The website for the AIF360 toolkit is: <https://github.com/IBM/AIF360>. It includes installation instructions which we will encourage the audience to complete before the course.

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